

Experiment 2: Water Jug Problem

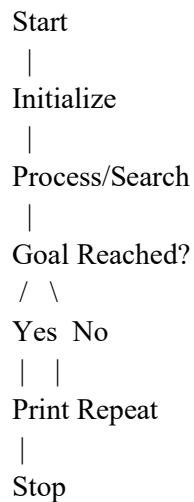
Aim

To implement and solve the Water Jug Problem using Artificial Intelligence techniques.

Algorithm

1. Start
2. Initialize the problem state
3. Apply valid operations/search
4. Repeat until goal state is reached
5. Display solution
6. Stop

Flowchart



Objective

- Understand state-space search.
- Implement AI search technique.
- Obtain the required goal state.

Program

```
from collections import deque

def water_jug(jug1, jug2, target):

    visited = set()

    queue = deque([(0, 0)])

    while queue:

        x, y = queue.popleft()
```

```

if (x, y) in visited:
    continue

print((x, y))
visited.add((x, y))

if x == target or y == target:
    print("Target Reached")
    return

next_states = [
    (jug1, y),          # Fill Jug 1
    (x, jug2),          # Fill Jug 2
    (0, y),             # Empty Jug 1
    (x, 0),             # Empty Jug 2
    (max(0, x-(jug2-y)), min(jug2, x+y)), # Pour Jug 1 → Jug 2
    (min(jug1, x+y), max(0, y-(jug1-x))) # Pour Jug 2 → Jug 1
]

for state in next_states:
    if state not in visited:
        queue.append(state)

water_jug(4, 3, 2)

```

Output



The screenshot shows a code editor interface with a file named 'main.py'. The code is a Python function 'water_jug' that uses a breadth-first search algorithm to find a path from (0,0) to a target (4,2) in a 2-jug water problem. The output panel on the right shows the sequence of states visited: (0, 0), (4, 0), (0, 3), (4, 3), (1, 3), (3, 0), (1, 0), (3, 3), (0, 1), (4, 2), followed by 'Target Reached' and '=== Code Execution Successful ==='.

```
main.py  [Full Screen] [Theme] [Share] [Run] Output

2 from collections import deque
3
4 def water_jug(jug1, jug2, target):
5     visited = set()
6     queue = deque([(0, 0)])
7
8     while queue:
9         x, y = queue.popleft()
10
11         if (x, y) in visited:
12             continue
13
14         print((x, y))
15         visited.add((x, y))
16
17         if x == target or y == target:
18             print("Target Reached")
19             return
```

Output

(0, 0)
(4, 0)
(0, 3)
(4, 3)
(1, 3)
(3, 0)
(1, 0)
(3, 3)
(0, 1)
(4, 2)
Target Reached
=== Code Execution Successful ===

Result

The program was executed successfully.

Conclusion

The program demonstrated the AI concept successfully and produced the expected result.